Vibration-Based Fault Detection and Classification in Unmanned Aerial Vehicles

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Abstract

Reliable detection of propulsion-system faults is essential for the safe and efficient operation of unmanned aerial vehicles (UAVs). This work investigates a vibration-based fault detection approach using inertial measurement data. The methodology involves signal preprocessing, frequency- domain analysis using the Fast Fourier Transform (FFT), Power Spectral Density (PSD) estimation, and extraction of statistical and spectral features. These features are used to distinguish healthy operation from selected propulsion fault conditions. The performance of the proposed approach is evaluated using appropriate classification metrics.

Index Terms

UAV, fault detection, vibration analysis, accelerometer, FFT, power spectral density, machine learning

I. Introduction

Unmanned Aerial Vehicles (UAVs) have a wide range of applications in today's world ranging from agricultural purposes all the way to military purposes. Therefore, it is important for UAVs to have reliable propulsion systems because even small mechanical problems might lead to unusual vibrations, inefficiency, or even inability to fly. Although traditional fault detection requires installing expensive external sensors or conducting a visual inspection, this paper proposes a solution for detecting mechanical problems in UAVs with built-in IMU accelerometers without adding any additional weight.

The proposed methodology is based on using vibration for the identification and classification of three typical UAV propulsion faults: damaged propellers, bearing problems, and motor misalignments. As a single parameter might be unreliable due to different flight modes, this approach involves analyzing the data obtained from accelerometers in all three axes (X, Y, and Z): time domain parameters (RMS acceleration) and fast Fourier transform or power spectral density in frequency domain. Preprocessing the signal and calculating several combined parameters (dominant frequencies, spectral peak magnitudes, frequency-band powers) allows for separating faulty and healthy flights of the UAV.

II. Related Work

- UAVs being increasingly used in today's world, also requires real-time fault diagnosis to ensure safety and reliability during operation.
- Vibration monitoring using accelerometers along with fault classifications algorithms helps us detect anomalies in flight behavior before instabilities even occur.
- According to previous studies, it was found that multi-resolution transform features given to deep neural networks achieve 91% accuracy for UAV fault classification.
- It was also found that when frequency variations were analyzed with wavelet scattering and long short-term memory (LSTM) models, they managed to achieve upto 99% accuracy in propeller damage detection.
- According to another paper, decision tree and support vector machine algorithms are highly efficient for low processing power applications (It was possible to achieve 93.27% accuracy using 14 features and 98.21% accuracy using 64 features.
- Vibrations below 200 Hz can lead to structural deformations and screw loosening in UAV frames.
- The conditions in a UAV which leads to screw-loosening have been simulated and detected in lab environments by analyzing time-dependent vibration amplitudes.
- However, solely relying on the amplitude of vibration as the distinguishing feature for fault-detection can be unreliable due to shifts in the UAV's natural frequencies at different speeds.
- Attempts have been made to carry out real-time onboard diagnosis using accelerometer and gyroscope data with PCA and ANN/SVM models and FPGA-based deep learning networks.

III. Dataset and Fault Conditions

A. Simulated Dataset

- Accelerometer datasets were generated via Monte Carlo simulation to assess the validity of our classification algorithm.
- The simulation produced one-second data windows sampled at 8 kHz, operating at a nominal speed of 6000 RPM.
- The simulated fault classes included a healthy baseline, mass imbalance, shaft misalignment, and bearing defects.
- To test noise robustness, the simulation generated 30 data windows per fault class across varying Gaussian sensor noise levels.

B. External Recorded Dataset

- A DJI Mini 2 Combo quadcopter operating in a steady hover mode at a fixed altitude of 1.2 meters was utilized as a basis for an external validation dataset.
- Vibration measurements were captured using an ADXL335 triaxial analog accelerometer mounted at the geometric center of the UAV, where the four rotor arms intersect.
- For the external dataset, data acquisition was managed by a DAQ-6008 USB device.
- The physical fault conditions tested included a healthy state, mass removal on one blade to simulate mechanical wear, and weight unbalance induced by affixing sticky tape to one blade.
- Each flight session lasted 7 minutes (420 seconds), generating approximately 400,000 raw vibration samples per flight case.
- The recorded time-domain signals were fed into Excel files containing time, X, Y, and Z-axis acceleration data.
- The sampling rate for this external recorded dataset was 1024 Hz, which approximately yielded 390, 1-second windows per data file.

IV. Methodology

A. Signal Preprocessing

Raw discrete accelerometer signals are initially mean-centered . A 4th-order Butterworth high-pass filter with a cutoff frequency of $f_c = 10$ Hz is applied zero-phase to all axes . This step removes the macro-dynamic wander of the rigid body and the stationary 1g gravity bias on the vertical axis prior to spectral decomposition . Finally, the time-domain signal is multiplied by a Hann window to limit the spectral smearing caused by discontinuities at the window boundaries .

B. Mathematical Models

To model how localized mechanical faults propagate to the flight controller's accelerometer, the UAV chassis is approximated as a Linear Time-Invariant (LTI) mass-spring-damper system. The structural Accelerance transfer function $H_x(\omega)$ is parameterized using the equivalent mass (m_{eq}), structural damping (c_x), and stiffness (k_x) of the chassis:

$$H_x(\omega) = \frac{-\omega^2}{-m_{eq}\omega^2 + jc_x\omega + k_x} \tag{1}$$

1) Propeller Mass Imbalance: Propeller mass imbalance behaves as a point mass $m_{u,i}$ located at a radial distance r_i for a given motor i. As the rotor spins at an angular velocity $\Omega_i = \frac{2\pi \cdot RPM_i}{60}$, it generates a continuous, rotating centrifugal force. The total physical forcing function on the horizontal axis is the sum of the individual centrifugal unbalance forces from all N motors:

$$F_x(t) = \sum_{i=1}^{N} m_{u,i} r_i \Omega_i^2 \sin(\Omega_i t + \phi_i) \tag{2}$$

The simulated accelerometer reading $a_x(t)$ is the convolution of this continuous force with the drone arm's structural impulse response $h_x(t)$, superimposed with the parameterized Gaussian sensor noise $\eta_x(t)$ (with variance σ^2):

$$a_x(t) = \left(\int_{-\infty}^{\infty} h_x(\tau) F_x(t-\tau) d\tau \right) + \eta_x(t) \tag{3}$$

The proposed model detects propeller blade damage using vibration signals measured along the X and Y axes of an accelerometer. The recorded signals are divided into non-overlapping 10-second decision windows. Each window is processed using the Fast Fourier Transform (FFT), power spectral density (PSD) estimation, spectral feature extraction, and logistic regression classification. Consecutive positive classifications are then used to determine whether a persistent damage alarm should be raised.

2) Rotating Shaft Misalignment: In direct-drive brushless motors, angular misalignment manifests as a rotating bent shaft with a permanent angular offset α. This tilt generates a dynamic couple imbalance dictated by the rotor's polar inertia I_p and transverse inertia I_t, which is reacted by bearings separated by a distance L_b. Additionally, the tilted propeller disk redirects a lateral aerodynamic thrust T component. The combined 1Ω radial forcing amplitude F_1 is:

$$F_1 = \frac{(I_p - I_t)\sin\alpha\cos\alpha \cdot \Omega^2}{L_b} + T\sin\alpha \tag{4}$$

Furthermore, anisotropic shaft stiffness (Δk) and axial bearing preload variations during rotation induce a 2Ω axial phenomenological forcing component F_2:

$$F_2 = \Delta k \cdot \alpha \tag{5}$$

For a multi-motor system, the integrated continuous forcing functions applied to the structural LTI system are:

$$F_x(t) = \sum_{i=1}^{N} F_{1,i}\sin(\Omega_i t + \phi_i) \tag{6}$$

$$F_z(t) = \sum_{i=1}^{N} F_{2,i}\sin(2\Omega_i t + \phi_{2,i}) \tag{7}$$

3) Bearing Imperfections: Unlike mass unbalance, bearing defects generate high-frequency periodic micro-collisions. The specific kinematic fault frequencies for inner ring (f_i), outer ring (f_o), and ball element (f_b) damage depend on the number of rollers Z, rolling element diameter d, pitch diameter D, and contact angle α:

$$f_i = \frac{Zf_r}{2}\left(1 + \frac{d}{D}\cos\alpha\right) \tag{8}$$

$$f_o = \frac{Zf_r}{2}\left(1 - \frac{d}{D}\cos\alpha\right) \tag{9}$$

$$f_b = \frac{Df_r}{2d}\left(1 - \frac{d^2}{D^2}\cos^2\alpha\right) \tag{10}$$

When a rolling element of mass m_b strikes a defect of width w_d with a kinematic constant c_k, the impact velocity $v_{impact} = c_k \cdot \Omega \cdot w_d$ converts kinetic energy into elastic strain energy. Assuming a Hertzian contact stiffness K_h, the peak impact force F_{max} scales linearly with the rotor speed and defect severity:

$$F_{max} = (c_k \cdot \Omega \cdot w_d)\sqrt{m_b K_h} \tag{11}$$

Projecting this 3D acoustic shock into the radial plane using the contact angle α and defect position angle β, the localized mechanical forcing function becomes a time-domain impulse train occurring at the fault interval $T_{fault} = 1/f_{fault}$ with a microscopic contact duration $\Delta t_{contact}$:

$$F_{bearing,x}(t) = (F_{max}\cos\alpha\cos\beta \cdot \Delta t_{contact})\sum_{n=0}^{\infty}\delta(t - nT_{fault}) \tag{12}$$

C. PSD and Regression

1) Signal Processing: The Discrete Fourier Transform (DFT) of a preprocessed and Hann-windowed frame $x_{q,m}[n]$ is:

$$X_{q,m}[k] = \sum_{n=0}^{N-1} x_{q,m}[n]\exp\left(-j\frac{2\pi kn}{N}\right) \tag{13}$$

The one-sided power spectral density (PSD) estimate for a single frame is computed as:

$$P_{q,m}[k] = \frac{c_k|X_{q,m}[k]|^2}{f_s\sum_{n=0}^{N-1} w^2[n]} \tag{14}$$

2) Class-Weighted Logistic Regression: The linear score for a scaled feature vector $\mathbf{z}_i$ and the resulting logistic damage score s_i:

$$l_i = \boldsymbol{\beta}^T \mathbf{z}_i + b \tag{15}$$

$$s_i = \frac{1}{1 + \exp(-l_i)} \tag{16}$$

The model minimizes the class-weighted logistic loss with an L2 penalty:

$$\mathcal{L}(\boldsymbol{\beta}, b) = \frac{1}{n_{tr}} \sum_{i=1}^{n_{tr}} \alpha_i [\ln(1 + \exp(l_i)) - y_i l_i] + \frac{\lambda}{2} \|\boldsymbol{\beta}\|_2^2 \tag{17}$$

D. Frequency-Domain Analysis

In the frequency domain, the output acceleration spectrum $A(\omega)$ is modeled as the input forcing spectrum multiplied by the chassis Accelerance Frequency Response Function (FRF) $H(\omega)$, plus the frequency-domain noise floor $\Gamma(\omega)$. Because random-phase noise magnitudes cannot be linearly added, the expected bin Signal-to-Noise Ratio (SNR) for a pure sinusoidal signal of amplitude A submerged in white noise with standard deviation σ, utilizing an N-point Fast Fourier Transform (FFT), is defined as :

$$SNR_{bin} = \left(\frac{A^2}{2\sigma^2} \right) \frac{N}{2}$$

Furthermore, localized bearing impacts excite high-frequency structural resonances, producing an Amplitude Modulated (AM) signal . To isolate these impacts, the signal undergoes Hilbert Transform analytic demodulation to extract the amplitude envelope $e(t)$ before a secondary FFT is applied to obtain the envelope spectrum.

E. Feature Extraction

To ensure scale invariance and independence from raw ADC counts or specific acceleration units, all spectral thresholding relies strictly on Peak-to-Floor Ratios (PFR) . The fundamental rotation speed (f_0) is estimated natively from the combined radial spectrum using a harmonic-weighted score function :

$$Score_j = P_r(j) + 0.5P_m(2j) + 0.25P_m(3j)$$

where $P_r(j)$ is the PFR of the j-th bin and $P_m(j)$ is the maximum-filtered PFR .

Impulsiveness is quantified using excess kurtosis (Fisher's definition) on fractional-bandwidth band-pass filters to detect the "tailedness" of the amplitude distribution relative to a normal distribution :

$$K = \frac{\frac{1}{N} \sum_{i=1}^{N} (x_i - \bar{x})^4}{\sigma^4} - 3$$

To evaluate misalignment, an axial-to-radial ratio is computed . To prevent random noise from artificially inflating spectral peaks at low SNRs, the expected noise power variance is algebraically subtracted from the measured spectral bin magnitude . The Z-axis noise follows a Rayleigh distribution ($E[A^2] \approx 1.44 \cdot median^2$), while the combined radial spectrum follows a Chi-Square distribution ($E[A^2] \approx 1.19 \cdot median^2$) .

F. Fault Classification

The classification algorithm operates directly on the extracted features via a deterministic decision tree without requiring external RPM telemetry .

1) **Bearing Defect:** The algorithm checks the envelope spectrum for a fundamental peak and at least one integer harmonic with a PFR > 4.0 . If found, and the excess kurtosis of the selected band exceeds 0.3, the system classifies a Bearing Defect .
2) **Healthy Baseline:** If no impulsive bearing fault is detected, the algorithm checks the radial 1Ω PFR . If this value is < 3.5, the system defaults to the Healthy Baseline, representing normal mechanical and aerodynamic noise .
3) **Shaft Misalignment vs. Mass Imbalance:** If a valid radial 1Ω peak is present, the noise-bias-subtracted ratio of the Z-axis 2Ω harmonic against the radial 1Ω fundamental is calculated . If this ratio exceeds 1.0, the signature exhibits structural axial forcing characteristic of Shaft Misalignment . Otherwise, the purely radial centrifugal forcing is classified as Mass Imbalance.

Based on experimental Monte Carlo simulations evaluating performance across varying Gaussian sensor noise levels (σ), the classifier is highly robust under standard conditions. The algorithm achieved a 1.00 classification accuracy for healthy, imbalance, and misalignment windows up to a noise level of $\sigma = 0.30\ m/s^2$ (corresponding to a theoretical bin SNR of 16.3 dB). Bearing defect detection maintained a 0.80 accuracy rate at this same noise threshold. At higher extreme noise levels ($\sigma \geq 0.50\ m/s^2$), the overall mean confidence score drops gracefully, with missed detections failing safely into the Healthy Baseline rather than generating erratic false alarms.

V. Results and Discussion

A. Propeller Mass Imbalance

The data available in internet for multiaxial UAV vibration data was analysed to compare healthy, damaged-propeller, and added-mass conditions. The dataset contains X, Y, and Z acceleration measurements recorded during hover using an ADXL335 accelerometer mounted at the geometric centre of a DJI Mini 2 quadcopter. The following plots were obtained:

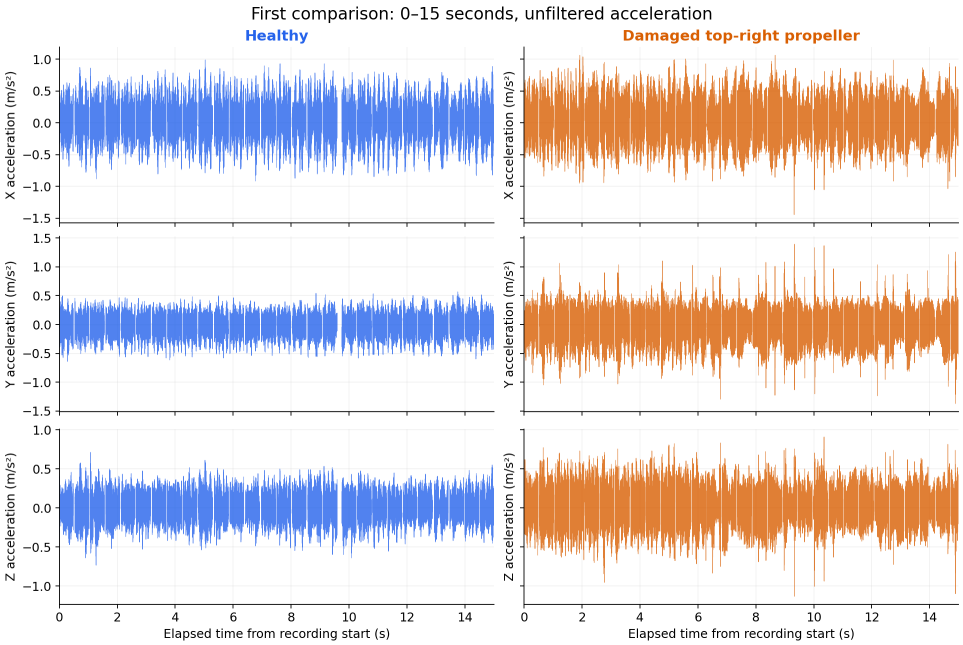


Fig. 1. Comparison of unfiltered triaxial acceleration signals along the X, Y, and Z axes for healthy and damaged-propeller UAV conditions over the first 15 seconds

Figure 1 compares the acceleration signals of the healthy and damaged-propeller conditions over the first 15 seconds. The damaged condition shows more noticeable fluctuations and occasional higher-amplitude peaks, particularly along the Y and Z axes. These differences suggest that propeller damage affects the UAV's vibration response.

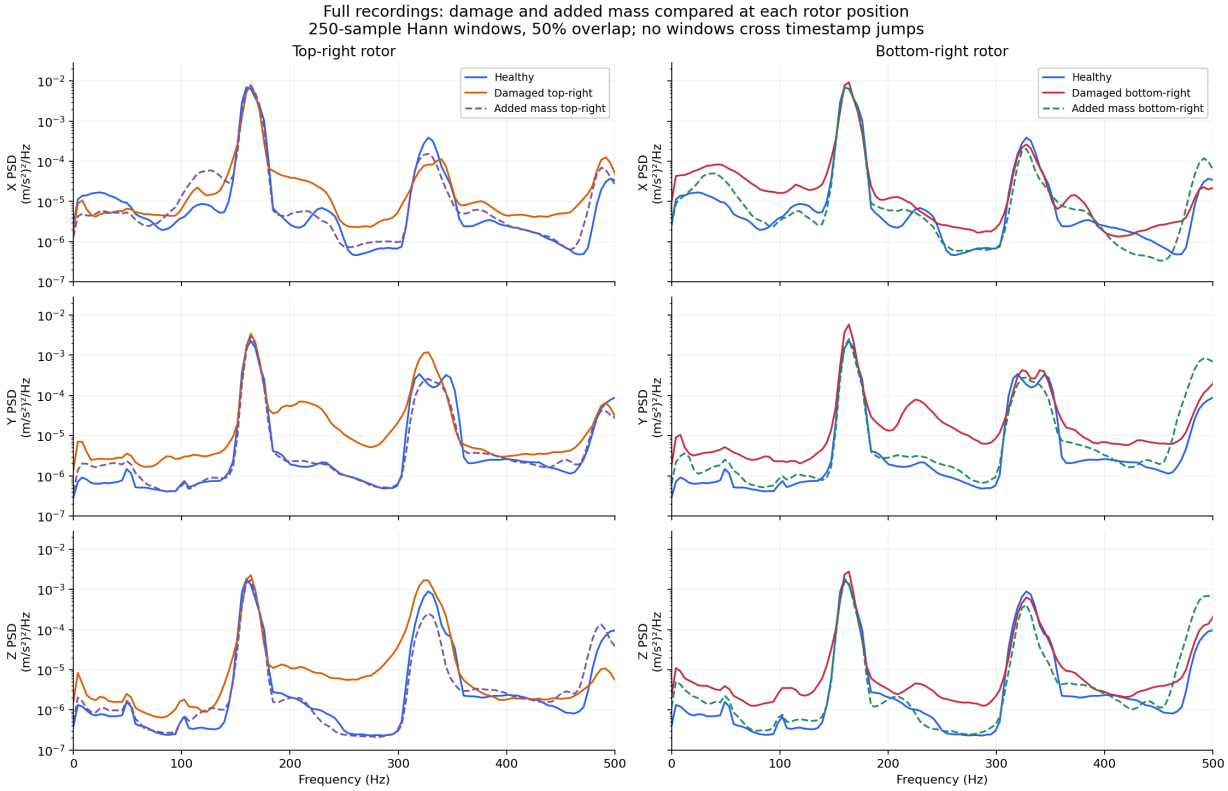


Fig. 2. Comparison of averaged power spectral density (PSD) across the X, Y, and Z axes for healthy, damaged-propeller, and added-mass conditions at the top-right and bottom-right rotor positions.

Figure 2 compares the vibration spectra for the two rotor positions under healthy, damaged-propeller, and added-mass conditions. A dominant peak appears around 160–165 Hz across all configurations, while additional peaks are observed near 320–345 Hz.

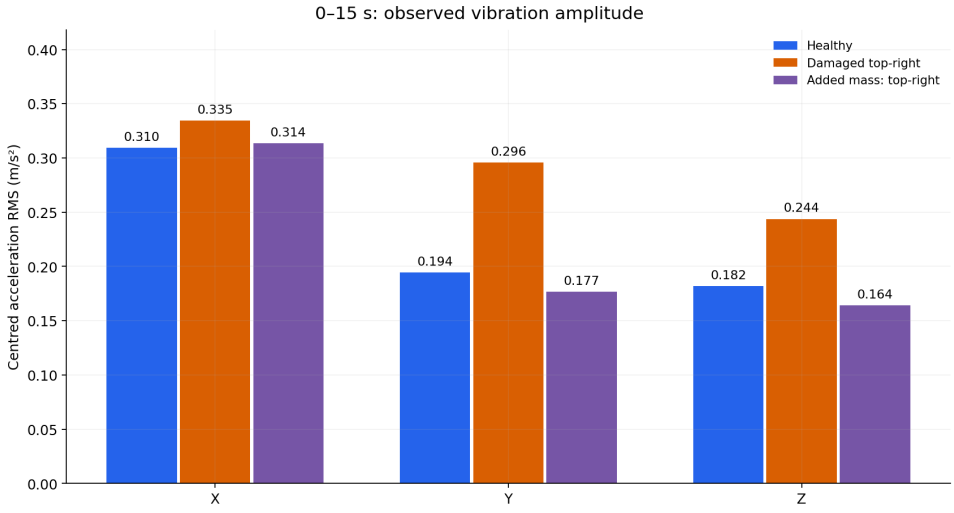


Fig. 3. Comparison of centred RMS acceleration along the X, Y, and Z axes under healthy, damaged-propeller, and added-mass conditions over the first 15 seconds.

Figure 3 compares the vibration levels across the three axes for the different propeller conditions. The damaged-propeller condition shows higher RMS values along all three axes, with the largest increase along the Y axis. In contrast, the added-mass condition produces RMS values close to the healthy case along X, but lower values along Y and Z.

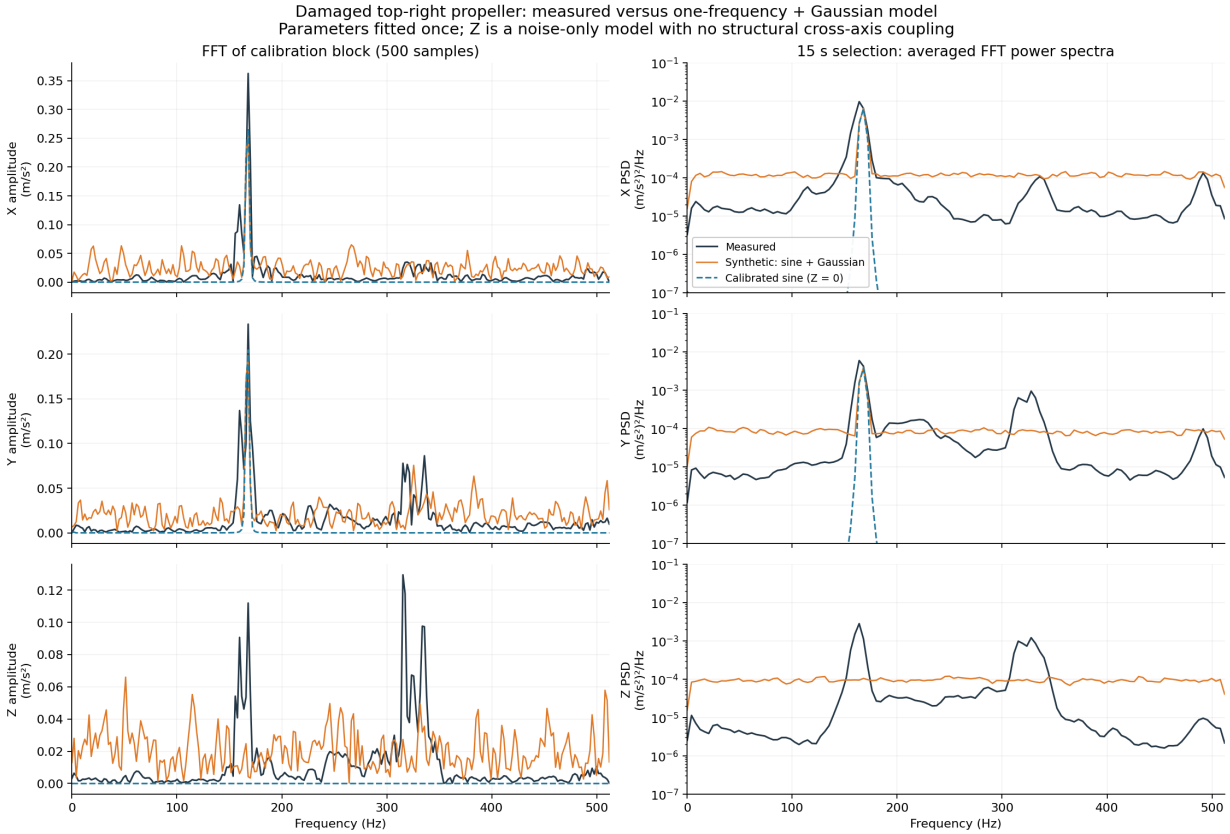


Fig. 4. Comparison of measured and synthetic acceleration spectra for the damaged top-right propeller along the X, Y, and Z axes, including the calibrated single-frequency model.

Figure 4 compares the measured acceleration spectra with the synthetic model for the damaged top-right propeller. The measured signals show several distinct peaks, particularly around 167 Hz and 320–340 Hz, whereas the synthetic model mainly represents the dominant frequency component with a relatively uniform noise background.

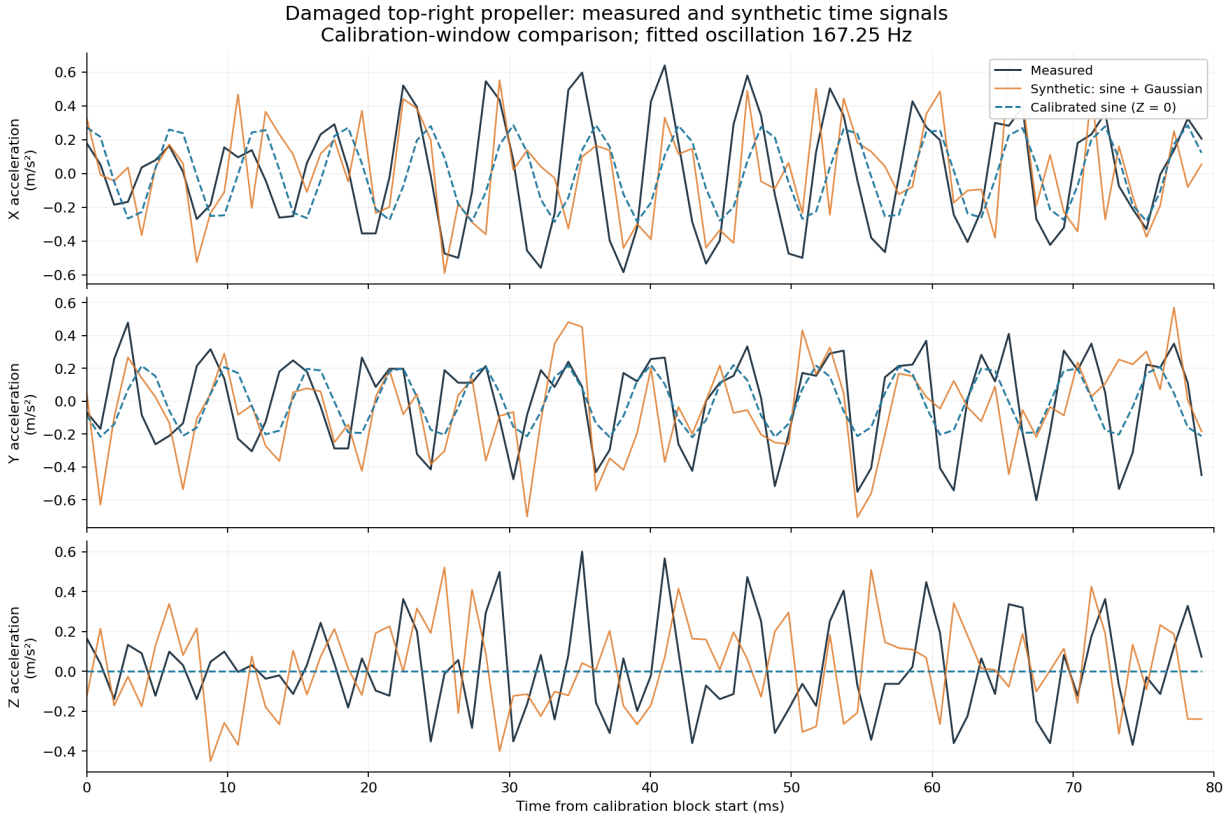


Fig. 5. Comparison of measured and synthetic acceleration signals along the X, Y, and Z axes for the damaged top-right propeller.

Figure 5 compares the measured and synthetic acceleration signals over an 80 ms interval. The synthetic model captures the approximate periodic behaviour of the X and Y signals, but noticeable differences remain in the amplitude and shape of the oscillations.

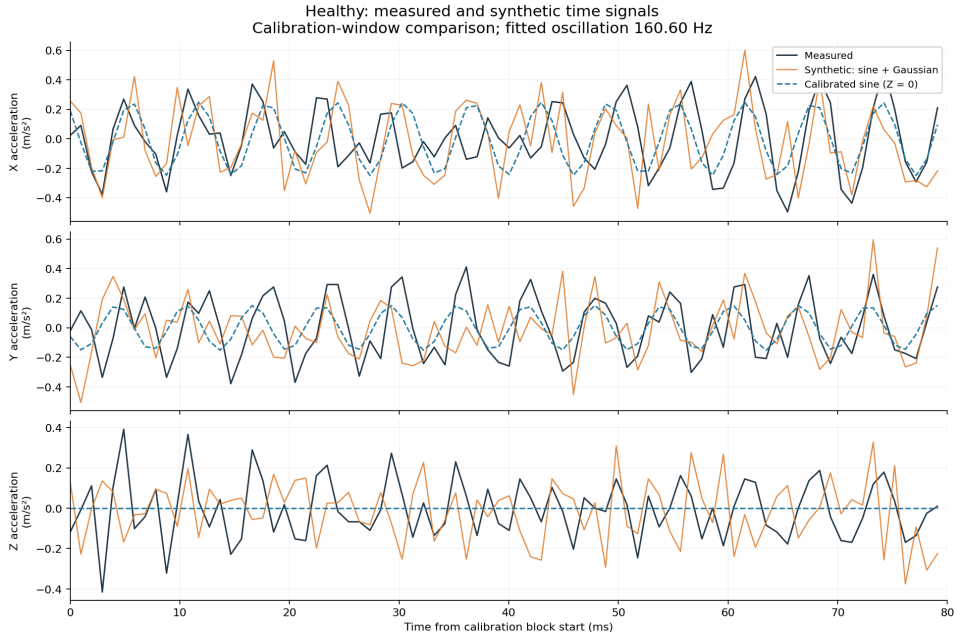


Fig. 6. Comparison of measured and synthetic acceleration signals along the X, Y, and Z axes for the healthy propeller.

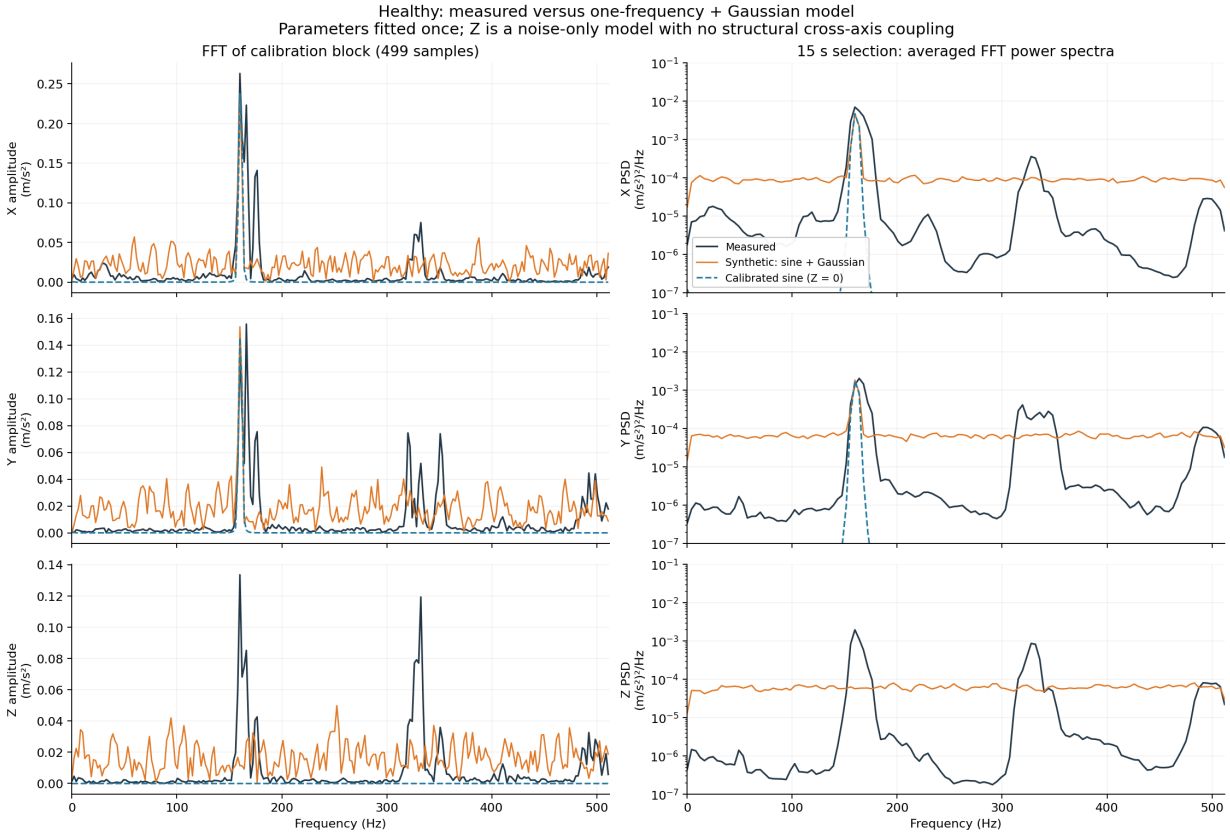


Fig. 7. Comparison of measured and synthetic acceleration spectra for the healthy propeller along the X, Y, and Z axes, including the calibrated single-frequency model.

As it can be observed from all these plots that it shows difference when plotted for damaged propeller and healthy propeller but not an appropriate method for detecting damage. Therefore a Machine Learning model is developed to detect damage, Model was trained from data of PSD of healthy propeller of a certain time interval which estimated baseline for detecting damage. The formulation for this is mentioned in the methodology section. Data of two different drones available on internet were analysed and two different base lines were obtained for both.

1) DJI Mini 2 quadcopter: For the DJI Mini 2 quadcopter whose plots are shown above also, the following plots and results were obtained:

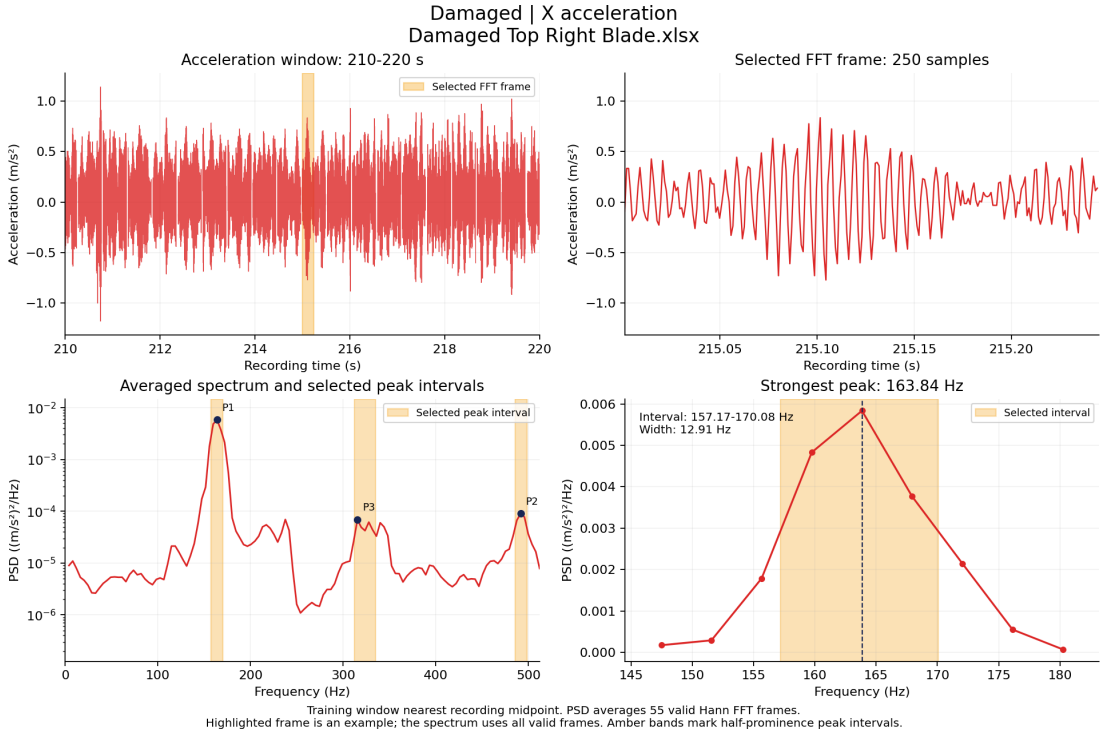


Fig. 8. Selection of a training window of X acceleration and identification of dominant frequency peaks using PSD for Damaged Propeller

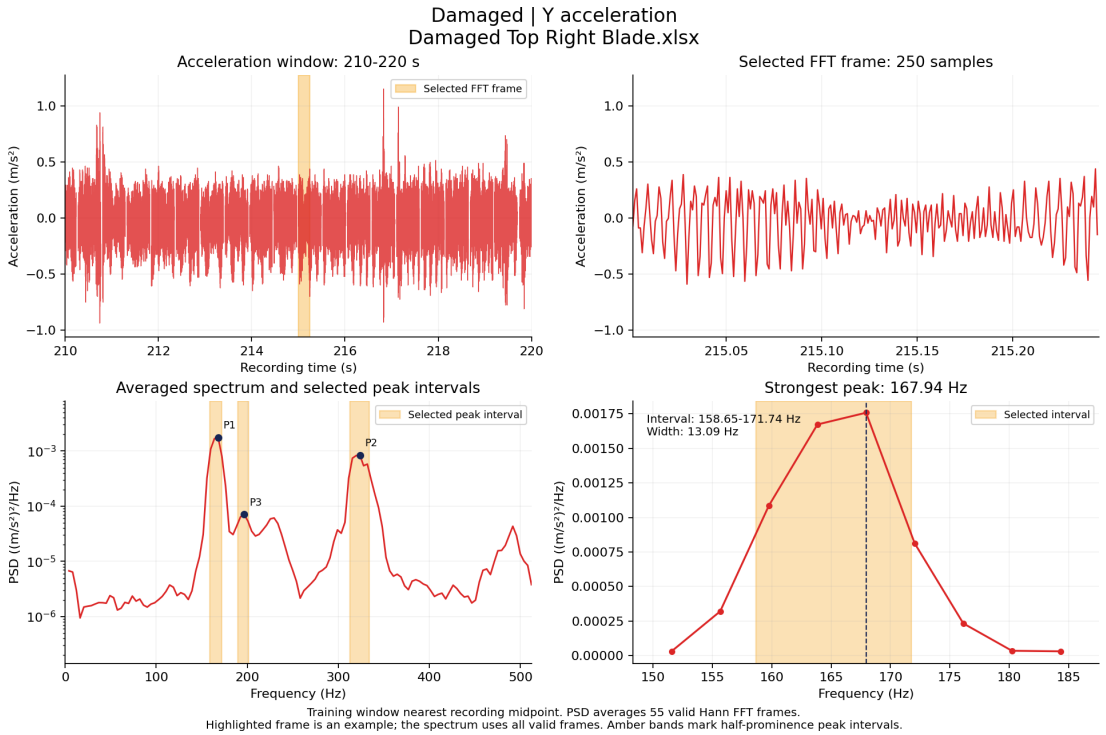


Fig. 9. Selection of a training window of Y acceleration and identification of dominant frequency peaks using PSD for Damaged Propeller

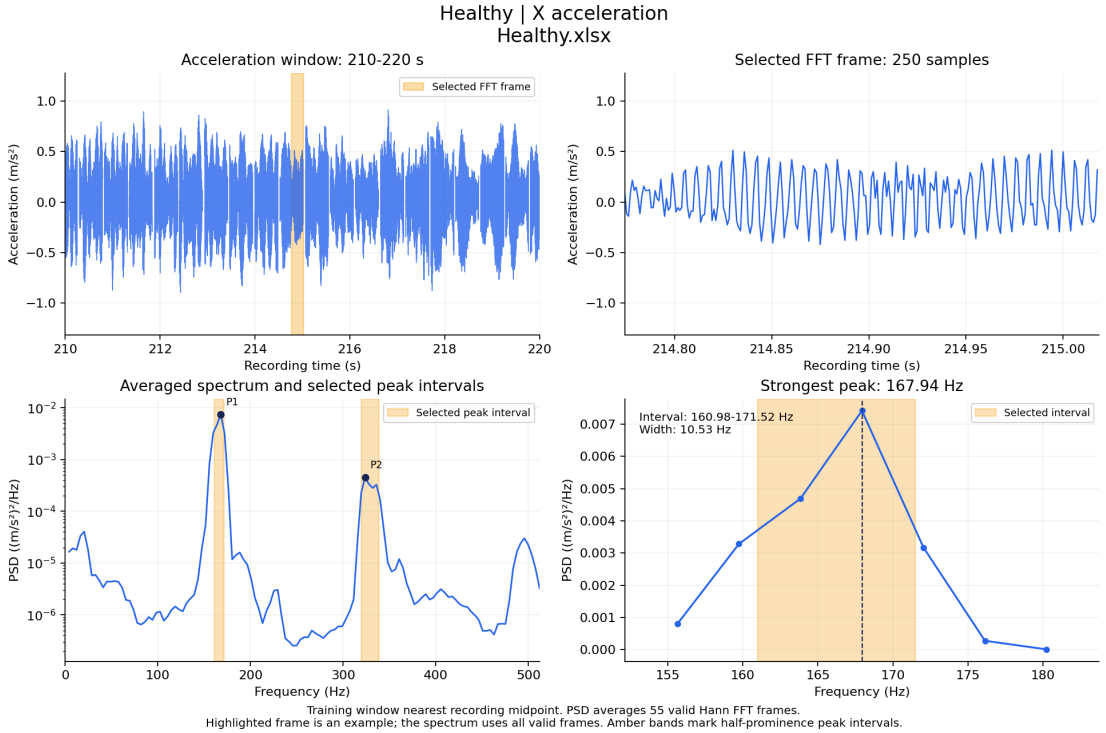


Fig. 10. Selection of a training window of X acceleration and identification of dominant frequency peaks using PSD for healthy Propeller

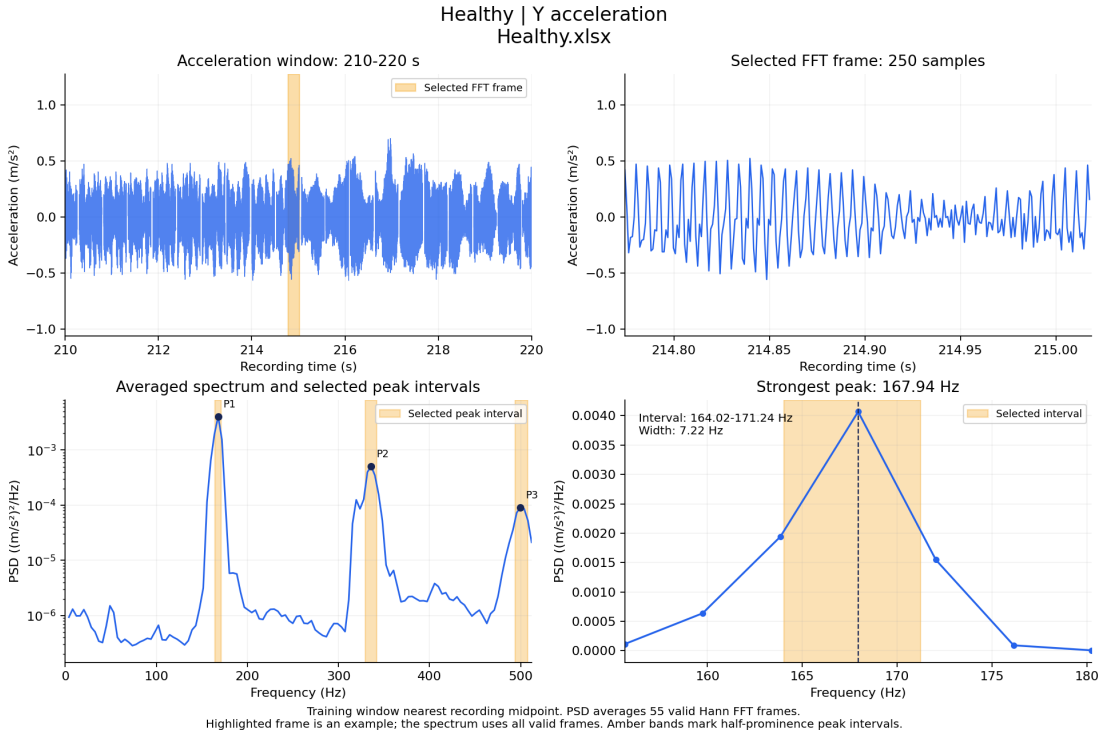


Fig. 11. Selection of a training window of Y acceleration and identification of dominant frequency peaks using PSD for healthy Propeller

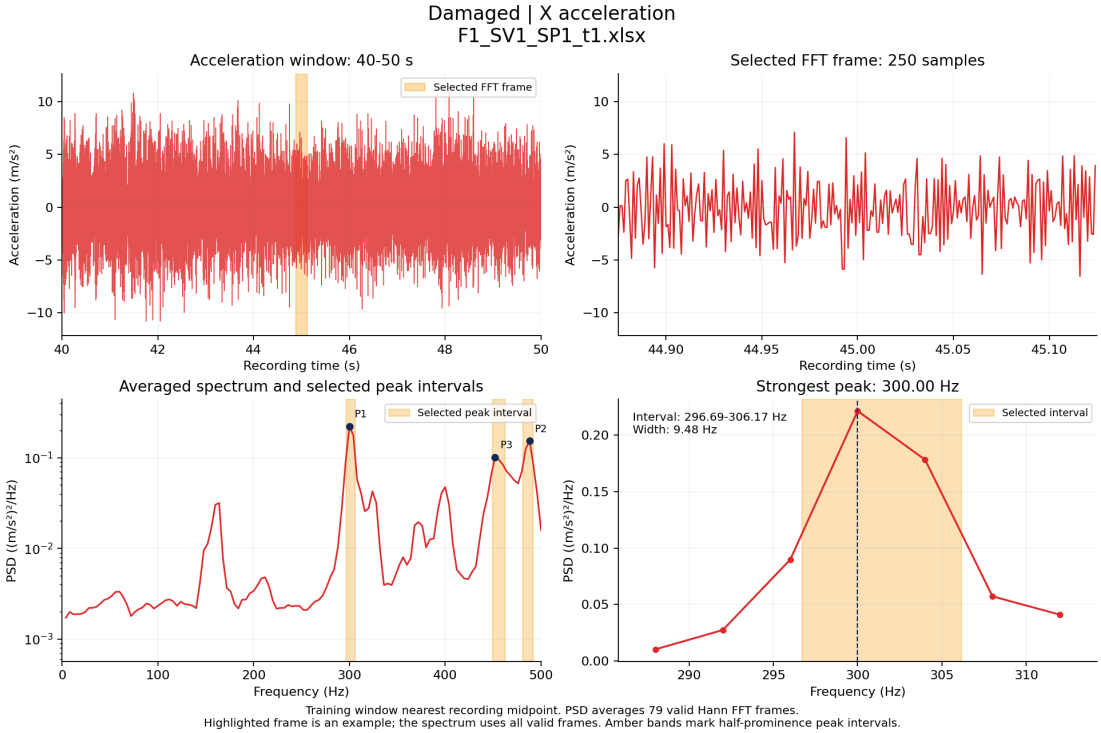


Fig. 13. Selection of a training window of X acceleration and identification of dominant frequency peaks using PSD for Damaged Propeller

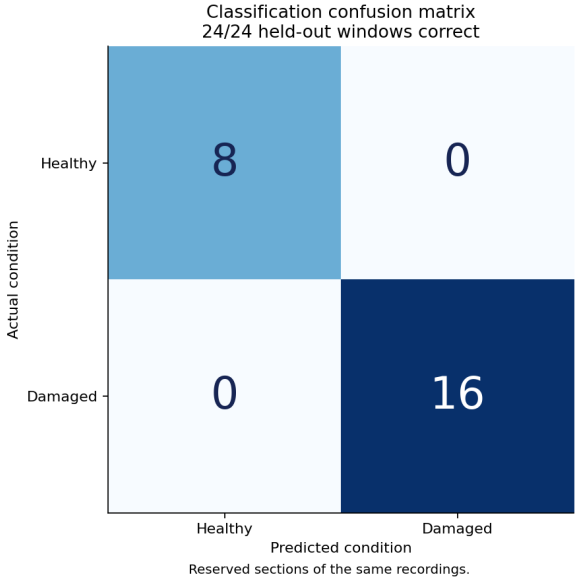


Fig. 12. Confusion Matrix for model of DJI Mini 2 quadcopter

Figure 12 shows the confusion matrix obtained from the machine learning model for detecting propeller damage. The model correctly classified all 24 held-out windows, including 8 healthy and 16 damaged windows, achieving an accuracy of 100% on this evaluation set. No false positives or false negatives were observed. Although for this drone dataset available was very limited so this verification may not be very proper.

2) Second Drone: The DronePropA dataset contains flight data collected from 130 indoor UAV flights. It includes both healthy propellers and damaged propellers with three types of faults: edge cuts, cracks, and surface cuts, each tested at three severity levels. The experiments cover five different flight trajectories, including cross-shaped and square-shaped movements, vertical climbs, and yawing manoeuvres. The model wad developed to detect faulty propeller or healthy propeller.

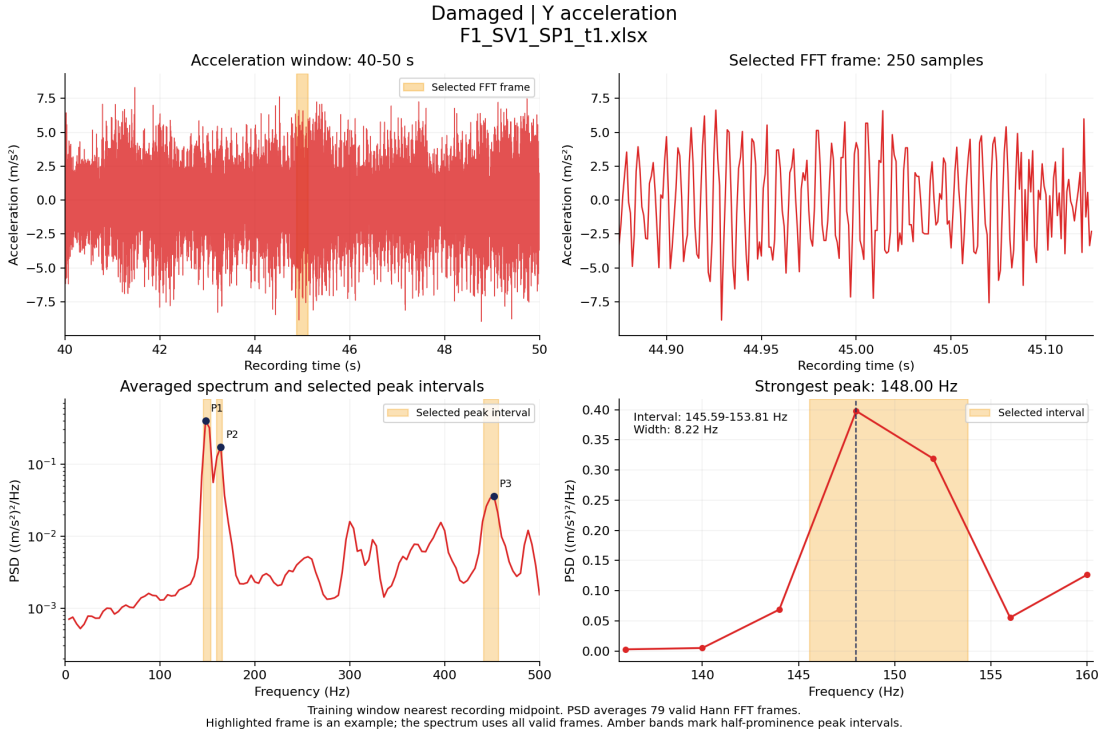


Fig. 14. Selection of a training window of Y acceleration and identification of dominant frequency peaks using PSD for Damaged Propeller

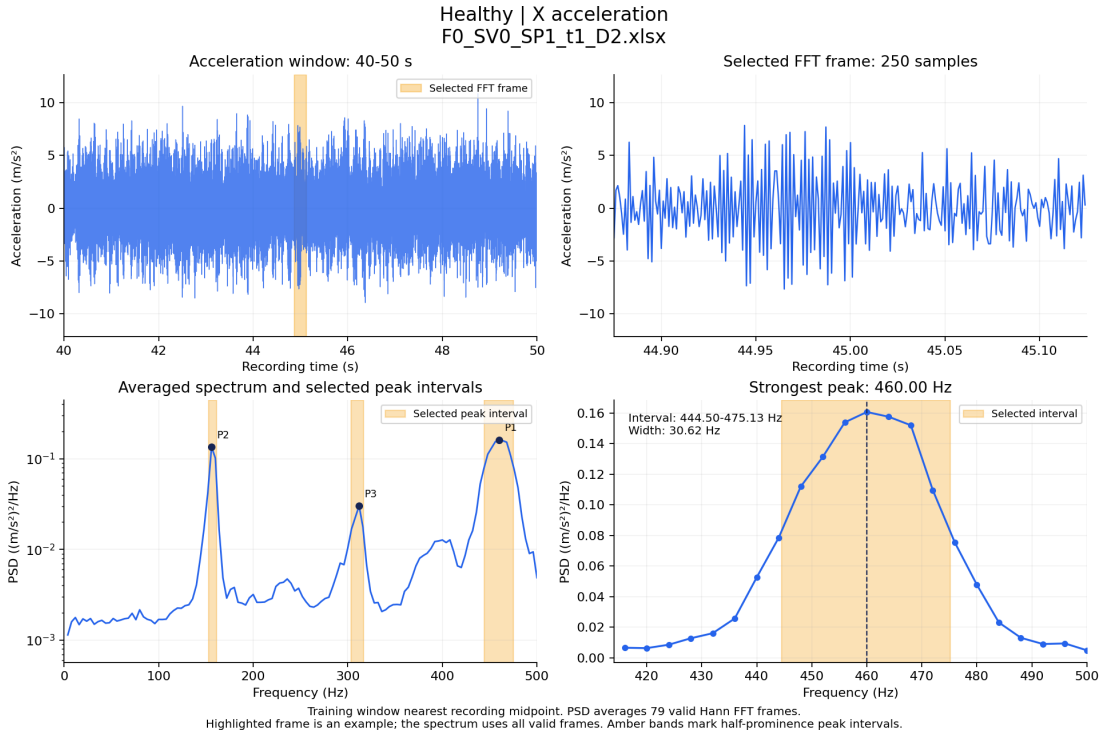


Fig. 15. Selection of a training window of X acceleration and identification of dominant frequency peaks using PSD for healthy Propeller

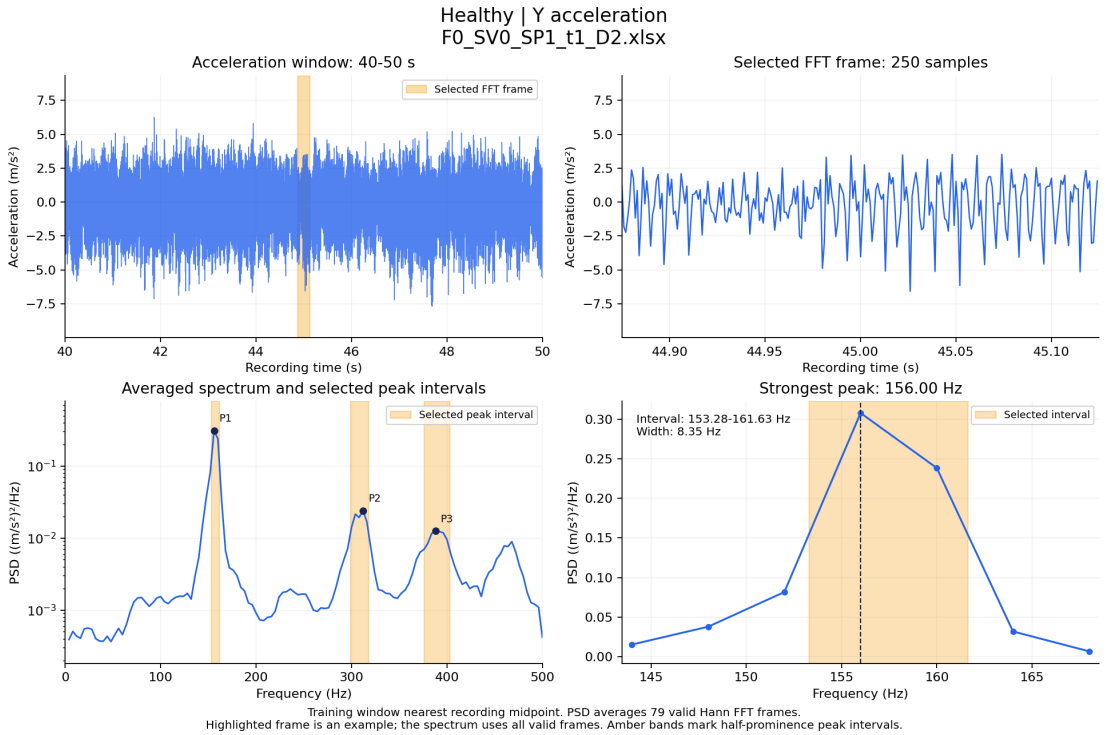


Fig. 16. Selection of a training window of Y acceleration and identification of dominant frequency peaks using PSD for healthy Propeller

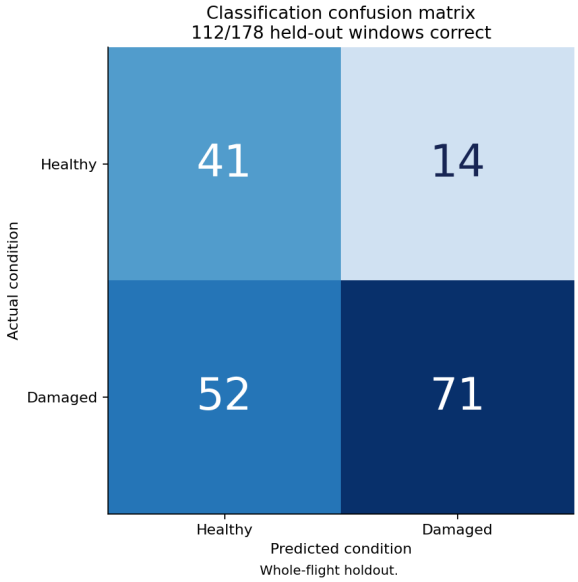


Fig. 17. Confusion Matrix

Figure 17 shows the model's performance on the held-out data from the DronePropA dataset. Out of 178 evaluated windows, 112 were classified correctly, giving an accuracy of 62.9%. The model correctly identified 41 healthy windows and 71 damaged windows. However, it also classified 14 healthy windows as damaged and failed to detect damage in 52 windows. These results show that the model can identify some damaged-propeller cases, but it still struggles to distinguish between some healthy and damaged conditions.

B. Observations of Second Model

The respective plots and figures related to the second signal processing algorithm (which covers all the three sources of error) are given below.

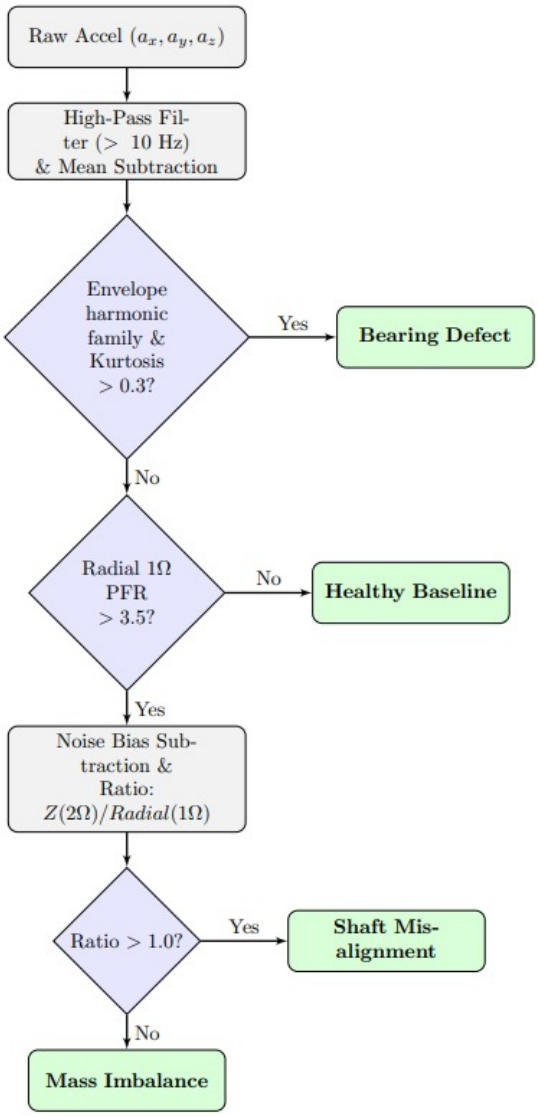


Fig. 18. Code Flowchart

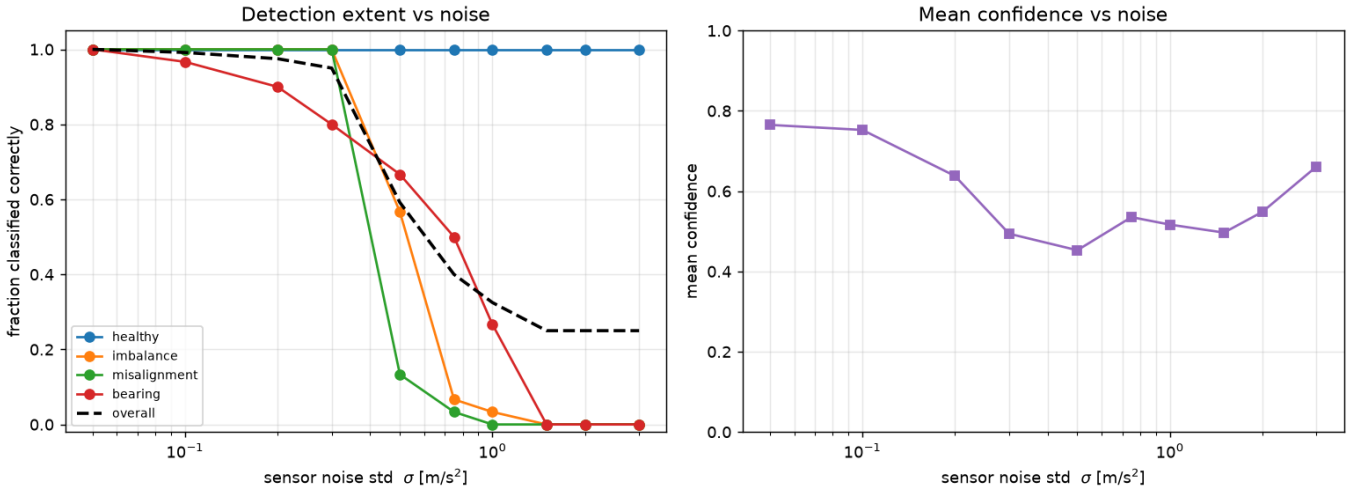


Fig. 19. These plots represent the variation of the fraction of defects correctly identified through classification with the standard deviation of the sensor noise (left plot) and the variation of mean confidence with respect to the sensor noise input (right)

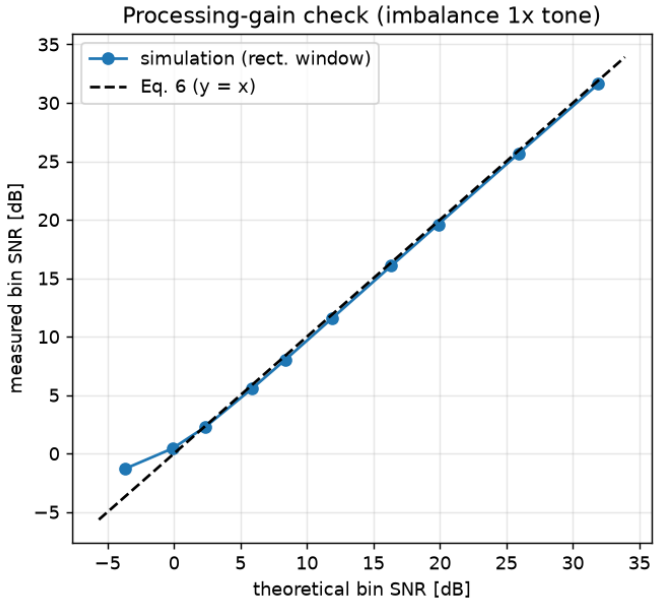


Fig. 20. This plot represents the variation of the measured bin signal noise ratio with the theoretical bin signal noise ratio.

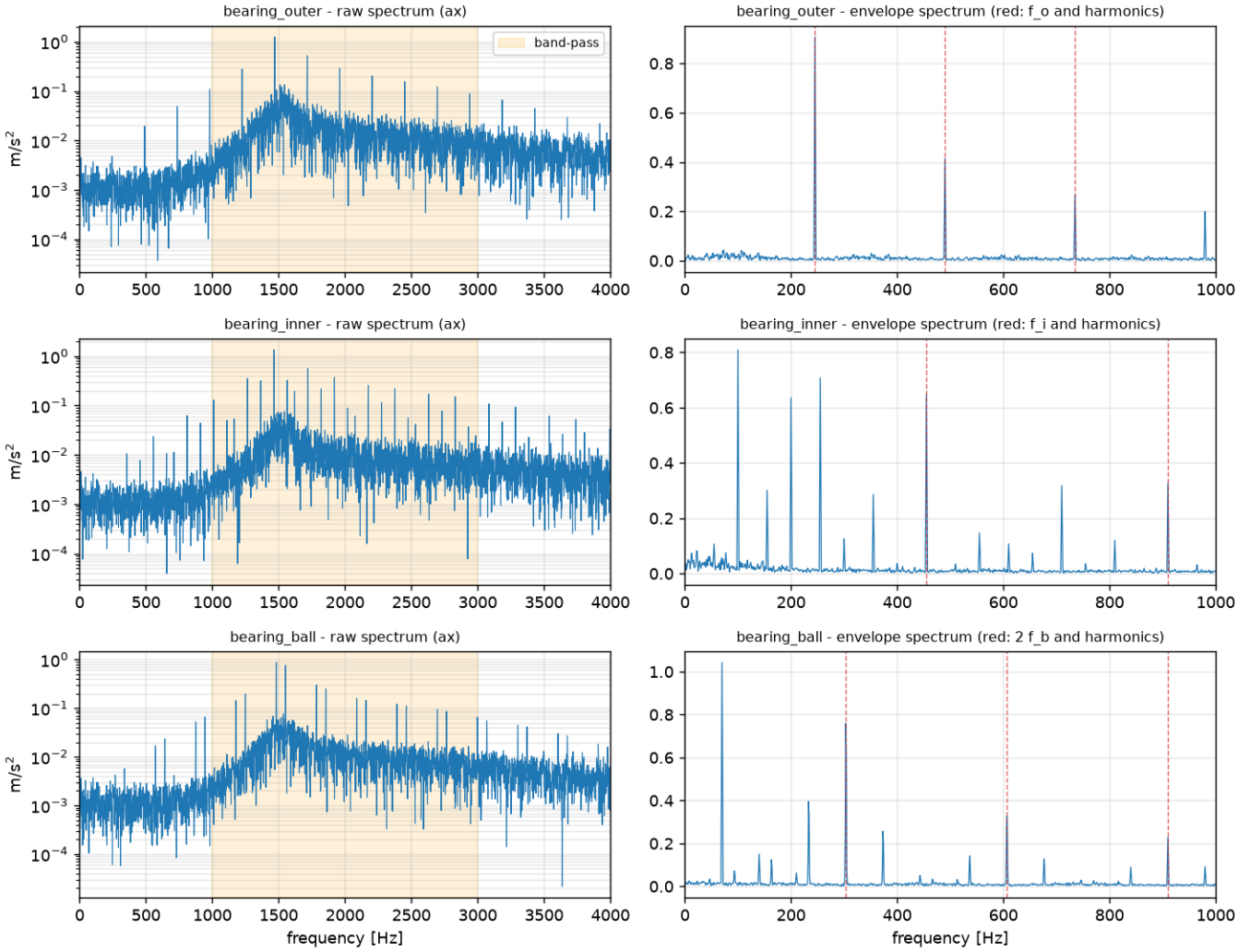


Fig. 21. These plots represent the acceleration-frequency spectra for the bearing defect case

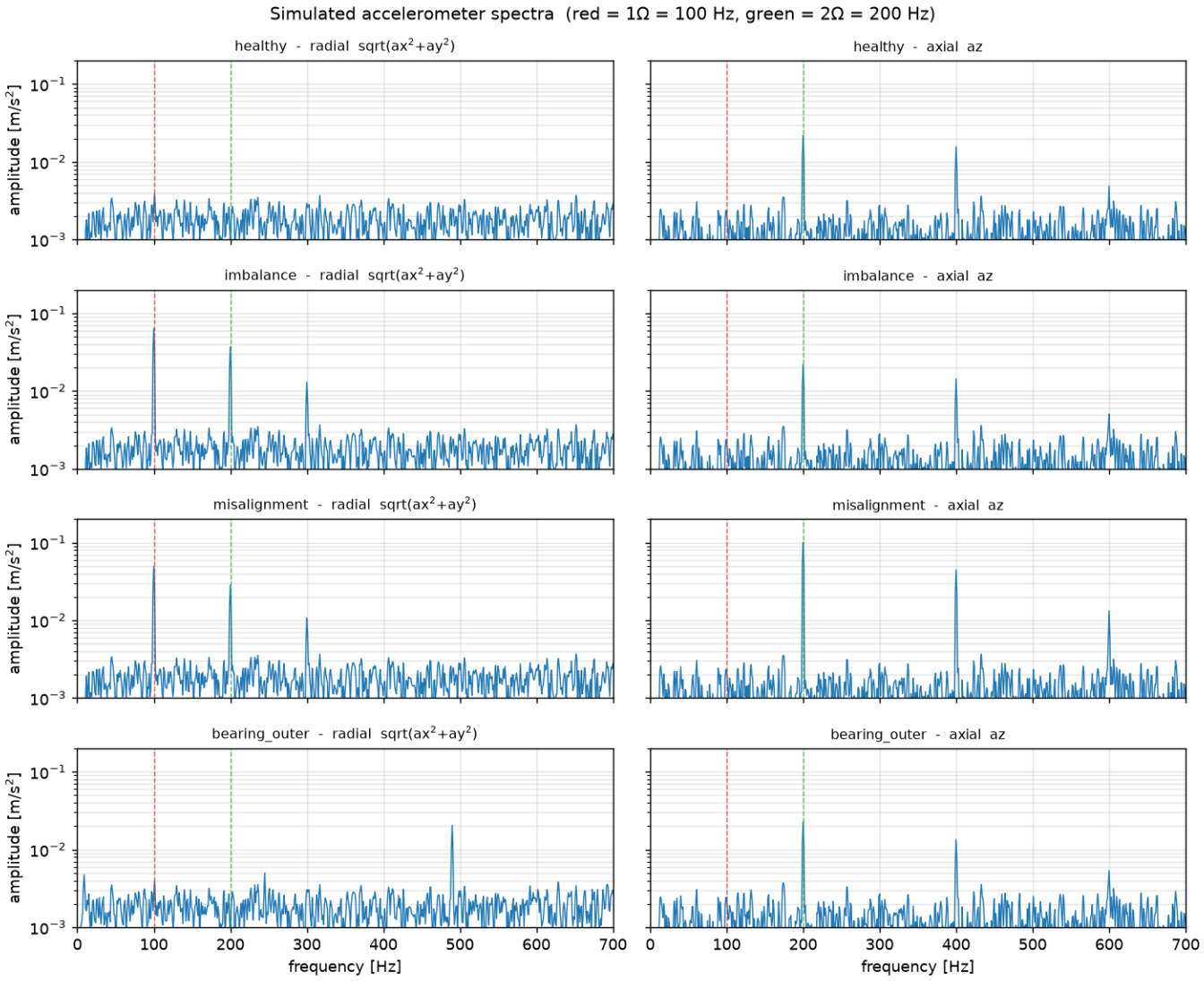


Fig. 22. These plots represent the acceleration-frequency spectra for the mass imbalance case

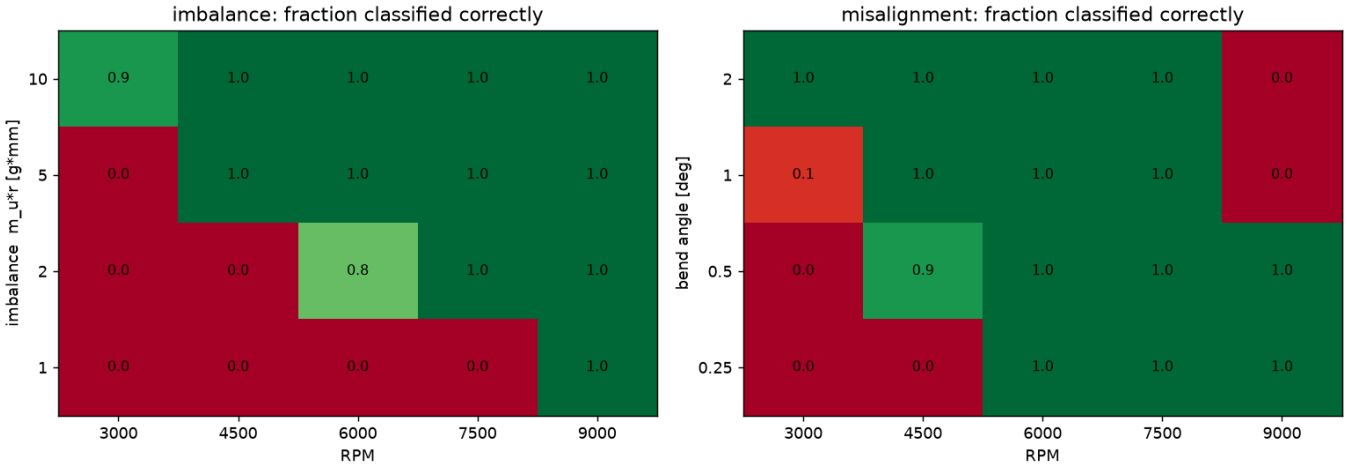


Fig. 23. The figure below represents the variation of the fraction of defects identified correctly with the operating RPM

Classification Decision Flowchart:

VI. Conclusion and Future Work

Our developed equations and fault detection and classification algorithms have been stress tested with various accelerometer data set available online with satisfactory confidence scores. Though there lies some degree of uncertainty and specifically that uncertainty scales with increasing noise, we still have managed to demonstrate the usefulness of signal conditioning and pre-processing on raw data to achieve our results.